Engineering Curves: Construction of Epicycloid

Rolling Circle Method with Tangent & Normal

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Problem

A generating circle of diameter 30 mm rolls externally without slipping on a directing circle of diameter 90 mm. Construct one complete epicycloid arch corresponding to $\angle AOB = 120^\circ$, divide the arch into 12 equal angular parts, and retain every construction element in all subsequent views.

Given Data and Calculations

Given data

$$
\begin{aligned}
\text{Diameter of directing circle} &= 90\ \text{mm},\\
R &= 45\ \text{mm},\\
\text{Diameter of generating circle} &= 30\ \text{mm},\\
r &= 15\ \text{mm},\\
\angle AOB &= 120^\circ,\\
\text{Let number of divisions} &= 12.
\end{aligned}
$$

Calculations

$$
\begin{aligned}
OC_0 &= R + r = 45 + 15 = 60\ \text{mm},\\
AC_0 &= r = 15\ \text{mm},\\
\Delta\theta &= \frac{120^\circ}{12} = 10^\circ.
\end{aligned}
$$

Generating-circle division

$$\text{Division angle} = \frac{360^\circ}{12} = 30^\circ.$$

Rolling relation

Each 10° centre step $\leftrightarrow$ 30° generating-circle step.

Step 1: Draw OA

$OA = 45$ mm.